

TAHA TARKHANI

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Education

University Of Ottawa

Expected Dec 2027

Bachelor of Applied Science in Software Engineering

Ottawa, Ontario

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Object Oriented Programming, Software Engineering, Software Design and Architecture, Algorithm Design and Analysis, Database Systems, Networks

Experience

National Defence Canada

May 2026 – Aug 2026

Software Engineer Intern (Returning)

Ottawa, ON

- Built Space-COP's **Portal Requests** dashboard card from scratch and a full-screen mode spanning **6** dashboard views, plus a reusable filter component adopted across **3** pages, in **React/TypeScript**.
- Rebuilt a sequential satellite tracking feed parser in **Java** using a bounded **ThreadPoolExecutor** and in-memory queue, cutting batch ingest time for 12K records from 6.5 minutes to 95 seconds, reducing it by **76%**.
- Accelerated release cycle by **45%** by designing and implementing automated **CI/CD** pipelines, enabling daily deployments while maintaining **99.7%** uptime for critical defense operations.
- Improved code reliability by increasing test coverage from **32%** to **78%** through comprehensive **JUnit** and integration tests, preventing mission-critical bugs before production.

National Defence Canada

Sep 2025 – Dec 2025

Software Engineer Intern

Ottawa, ON

- Optimized **PostgreSQL** query performance for satellite tracking data with composite indexes and monthly table partitioning, cutting peak query execution time from 1,800ms to 210ms, reducing it by **88%**.
- Migrated the team off manual deployments by building a multi-stage **GitLab CI/Docker** pipeline, cutting deployment turnaround from 25 minutes to an automated 8-minute process, reducing it by **68%**.
- Expanded ingest module test coverage from **18%** to **62%** using **JUnit 5** and **Testcontainers** for PostgreSQL integration tests, catching coordinate-wraparound bugs before staging.

National Research Council Canada

Jan 2025 – Apr 2025

Data Analyst Intern

Ottawa, ON

- Automated data processing with **Python** scripts for scientific research datasets (**PostgreSQL/MySQL**), reducing manual data entry time by **70%** through **ETL** workflows at scale.
- Designed a data pipeline architecture across **24 PostgreSQL/MySQL** tables, refactoring slow analytical joins and indexing to cut daily query execution times from **42 min** to **8 min** for 15+ researchers across 3 analytics tools.
- Implemented schema-validation guards quarantining **3.4%** malformed records per ingest, reducing weekly pipeline rerun incidents from **8** to **1** and authoring runbooks that cut onboarding from **3 days** to under **4 hours**.

Projects

Request Tracer | *Java, Spring Boot, React, Docker, REST APIs*

- Built and verified end-to-end correlation ID propagation across **4** independent **Spring Boot** services (Gateway, Orders, Inventory, Log Aggregator), confirming one identical trace ID across all **6** logged events per request.
- Developed a **React** dashboard rendering live traces as a waterfall timeline, and containerized the full system as **5** Docker images via **Docker Compose**, with real end-to-end latency isolating a one-time **1565ms** cold-start from a stable **64-119ms** steady state.

Orbit Tracker | *Python, Flask, Skyfield, globe.gl, REST APIs*

- Engineered a Python service using **Skyfield** (SGP4 propagation) to track **30** live satellites from **CelesTrak's** public API, validating results against known orbital data (e.g., GOES-16 altitude within **0.02%** of its true geostationary orbit).
- Cached TLE data to cut repeat fetch times from **2.43s** to under **10ms**, reducing it by **99.6%**, and built a pass-prediction **REST API** finding **84** upcoming satellite passes in **106.6ms**, visualized on an interactive **3D** globe using **globe.gl**.

Cost Monitor | *Python, Flask, AWS Cost Explorer, Boto3, DynamoDB, Chart.js*

- Designed and validated an anomaly detection system against test data, correctly distinguishing a **2.9%** fluctuation (unflagged) from an **83.2%** cost spike (flagged) using configurable thresholds.
- Built an AWS cost monitoring dashboard in **Flask** with an in-memory caching layer achieving **sub-1ms** cache-hit response times, ingesting billing data via the **Cost Explorer API** into **DynamoDB**.

Technical Skills

Languages: Python, Java, JavaScript, C++, SQL, C#, TypeScript, HTML/CSS

Technologies & Tools: React, Spring Boot, Node.js, Flask, Docker, AWS, PostgreSQL, MySQL, Git, REST APIs, CI/CD, GitHub Actions, JUnit, Jest, Boto3, DynamoDB, Skyfield, Chart.js, globe.gl